

João F. Melo

Personal Website: joaofmelo.github.io | iNSPIRE Profile: inspirehep.net/authors/1762166

PROFESSIONAL CAREER

College Supervisor, University of Cambridge	Jan 2026 — Present
Postdoctoral Researcher, KU Leuven, Institute for Theoretical Physics	Jul 2023 — Jan 2025

HIGHER EDUCATION

PhD in Applied Mathematics and Theoretical Physics, DAMTP, University of Cambridge	Oct 2018 — May 2023
Thesis Title: The Many Scales of Quantum Fields and Gravity Supervisor: Jorge E. Santos	
Topics: Quantum Field Theory in Curved Spacetime, Renormalisation, Holography, Cosmology	
MASt in Applied Mathematics, University of Cambridge	Oct 2017 — Jun 2018
Result: Pass with Distinction, 90/100	
Exams Taken: Quantum Field Theory, Statistical Field Theory, String Theory, General Relativity, Black Holes	
Essay: T-Duality and Geometry of String Theory Advisor: Ron Reid-Edwards	
Graduate in Engineering Sciences – Engineering Physics, Instituto Superior Técnico, University of Lisbon	Sep 2014 — Jul 2017
Result: 18/20	
Core Topics: Theoretical Physics, Experimental Physics, Mathematics, Computing, Electrical Engineering	

TEACHING EXPERIENCE

College Supervisor, University of Cambridge	Jan 2026 — Present
• Courses: Applications of Quantum Mechanics, Electromagnetism, Statistical Physics • Instructor for small groups (1-3 undergraduate students) covering the examples sheets in the 2nd and 3rd year of the Mathematical Tripos	
Summer Camp Lecturer, AstroCamp, University of Porto	Aug 2025
• Course: Special Relativity: The Structure of Time and Space • Lecturer in an astrophysics summer camp (20 hours of lectures, 8 hours of practicals) for 10 school students from 15 to 17 years old • Wrote lecture notes, 2 example sheets, and 3 exams (4 hours total)	
Master's Thesis Supervisor, KU Leuven	Sep 2024 — Dec 2024
• Thesis Title: Quantum Field Theory in Curved Spacetime • Co-supervisor for the early stages for a Master's Thesis in Physics	
Bachelor's Project Supervisor, KU Leuven	Sep 2024 — Dec 2024
• Project Title: Black Hole Thermodynamics • Sole supervisor of 2 students for their “capstone” project in the 3rd year of the Bachelor of Physics	
Teaching Assistant, KU Leuven	Nov 2024
• Courses: Statistical Mechanics • Instructor for exercise sessions for ~ 20 students in the 3rd year of the Bachelor of Physics	
Lecturer, KU Leuven	Feb 2024 — Jun 2024
• Courses: Advanced Quantum Field Theory • Delivered one 2-hour lecture per week for 12 weeks for ~ 30 students in the Master of Physics • Wrote lecture notes and 3 graded homework assignments • Conducted the final oral examination consisting of short presentations on a project written by each student chosen from a set list	
Examples Classes Instructor, University of Cambridge	Jan 2019 — Jan 2022
• Courses: General Relativity, String Theory • Sole instructor for ~ 20 master's students in the 4th year of the Mathematical Tripos covering the example sheets	
Part III Preparatory Workshops Instructor, University of Cambridge	Oct 2019 — Oct 2021
• Courses: General Relativity • Interactive workshop for ~ 50 master's students covering pre-requisites for the 4th year of the Mathematical Tripos • Co-wrote notes with David Tong, these are still in use	
PhD Drop-Ins, University of Cambridge	Jan 2019 — Jun 2019
• Courses: General Relativity, String Theory, Statistical Field Theory • Office hours for master's students in the 4th year of the Mathematical Tripos	

ADMINISTRATIVE EXPERIENCE

Effective Field Theories Reading Group	Oct 2020 — Dec 2020
Founder and Organiser	
• Group for PhD students following the book “Effective Field Theories” by Andrew E. Blechman and Alexey A. Petrov	

Racism Discussion Group	Jun 2020 — Dec 2020
Co-Founder and Co-Organiser with Chandrima Ganguly, Hayley Macpherson, and Miren Radia	
• Group for staff and students to discuss the issues of racial and ethnic discrimination in academia	
Graduate Seminar	Oct 2019 — Jun 2020
Co-organiser with Maeve Madigan	
• Seminar for PhD students of the HEP and GR groups of DAMTP	
Pre-Seminar Meetings	Oct 2019 — Jun 2020
Founder and Co-Organiser with Filipe Miguel and Gonalo Regado	
• Meetings between PhD students and the speaker of the Quantum Fields & Strings and General Relativity DAMTP seminars	
Quantum Fields and Gravity Working Group	Mar 2019 — Mar 2020
Founder and Organiser	
• Group for PhD students at the intersection of the HEP and GR groups to share and discuss their research ideas	
International Physics Olympiads	Jun 2018
Liaison Officer	
• Prepared and planned the daily activities for the students with the organising committee	
• Coordinated a team of 10 guides each with their own delegation of 5 students	
• With a team of 2 other liaison officers (each with their own 10 guide team) coordinated the full logistics of ~ 150 students. Including meals, transportation, electronics storage, and conflict resolution	

OUTREACH

Demonstrating Experiments

Cambridge Hands-On Science	Oct 2018 — Mar 2020
• Target audience: school students from 8 to 12 years old at schools and science festivals	
NFIST - Physics Society of Instituto Superior Tcnico	Sep 2014 — Aug 2017
• Target audience:	
– School students from 14 to 18 years old at schools, youth correctional centres, and science festivals	
– General audience at music festivals and on the street	

Popular Science Writing

1. Clarke, P., Madigan M., **Melo, J. F.** Our Place in the Universe. *BlueSci*, 47 (2020)
2. **Melo, J. F.** Interview with David Tong (in Portuguese). *Pulsar*, 40 (2019)
3. **Melo, J. F.** Physics and Technology of the Telescope (in Portuguese). *Pulsar*, 38 (2017)
4. **Melo, J. F.** Physics and Technology of the Fridge (in Portuguese). *Pulsar*, 37 (2016)
5. **Melo, J. F.** Physics and Technology of the Television (in Portuguese). *Pulsar*, 36 (2016)
6. **Melo, J. F.** Physics and Technology of the Light Bulb (in Portuguese). *Pulsar*, 35 (2015)

SCIENTIFIC PUBLICATIONS

1. Valeixo Bento, B. & **Melo, J. F.** EFT & Species Scale: Friends or foes? *JHEP* **25**, 212. arXiv: 2501.08230 [hep-th] (2025).
2. **Melo, J. F.** *The Many Scales of Quantum Fields and Gravity* PhD thesis (Cambridge U., May 2023).
3. **Melo, J. F.** The propagator matrix reloaded. *SciPost Phys. Core* **6**, 019. arXiv: 2112.09119 [hep-th] (2023).
4. **Melo, J. F.** & Santos, J. E. Stringy corrections to the entropy of electrically charged supersymmetric black holes with $\text{AdS}_5 \times S^5$ asymptotics. *Phys. Rev. D* **103**, 066008. arXiv: 2007.06582 [hep-th] (2021).
5. **Melo, J. F.** & Santos, J. E. Developing local RG: quantum RG and BFSS. *JHEP* **05**, 063. arXiv: 1910.09559 [hep-th] (2020).
6. **Melo, J. F.** Introduction to Renormalisation. arXiv: 1909.11099 [hep-th] (Sept. 2019).

SCHOLARSHIPS AND AWARDS

PhD Studentship , Fundao para a Cincia e Tecnologia	Oct 2021 — Sep 2022
Smith-Knight and Rayleigh-Knight Prize , Faculty of Mathematics, University of Cambridge	Jan 2020
Vice-Chancellor’s Award , Cambridge Trust, University of Cambridge	Oct 2018 — Sep 2021
Getrude Mather Jackson Prize , Girton College, University of Cambridge	Jun 2018
M. T. Meyer Scholarship , Girton College, University of Cambridge	Jun 2018
Merit Scholarship , Instituto Superior Tcnico, University of Lisbon	Sep 2016 — Jul 2017

SHORT-TERM SCHOOLS

TASI , Black Holes, Quantum Information and Dualities	Jun 2021
AQFTUK/LMS , An Introduction to Quantum Field Theory in Curved Spacetime	Oct 2020
CERN , Supergravity, Strings and Gauge Theory	Feb 2019
LIP/CFTP , Particle and Astroparticle Physics	Feb 2017
University of Coimbra , Projecto Quark! – Preparation for the International Physics Olympiads	Jan 2014 — Jun 2014